

Beneath The Weight



Matilde W. S. Jensen & Luna Annabell L. Jensen

Luna Annabell Linné Jensen: 20213072
Matilde Wendy Stefania Jensen: 20216069
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Abstract

This report examines the intersection between environmental sustainability and art, through a dynamic sculpture *Beneath The Weight*, designed for the climate festival Aalborg2024. The Sculpture is created as a medium for evoking emotional responses and discourse on climate change and sustainable production. The research works with the questions of the role of art in the climate debate and material sustainability integrated in artistic practices.

Inspired by artworks like *Ice Watch* by Olafur Eliasson, the project uses theoretical frameworks of John Dewey and Christian Jantzen, to create a design method with a focus on sensory, emotional and intellectual engagement. The implementation involved creating a life-sized sculpture with human-like appearance using primarily recycled material, with an integration of sound and movement, responding to the intensity of information on climate change.

An analysis of walk along interviews with participants at the exhibition at Create (16 -17 of May 2024), with results of the sculptures ability to create sensory and emotional experiences, generating intellectual reflections on sustainability. The results suggest that dynamic sculptures can bridge gaps between art and life, making the environmental issues engaging and xx

The report concludes that the role of art can effectively create public engagement on environmental sustainability. As it can serve as a medium for reflection and discussion. The scope of the project can further be expanded, through cultural and social contexts and be improved by elements to enhance the impact on sustainable discourse.

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1. Introduction

Climate change is one of the most significant subjects in people's lives. Many fields delve into this subject, the artworld among them. More and more artists are working with climate change through their artworks, showcasing them in all sorts of places. The dynamic sculpture *Beneath The Weight* will be showcased at *Aalborg2024*. The prototype of the sculpture has been designed as a part of research on sustainability and art, with perspective on the emotional impact of climate change. Art has a unique capability to evoke emotions and create spaces for reflection and discourse.

This research investigates the potential in art, specifically dynamic sculpture, to create a public engagement with environmental sustainability. The project examines how the design and sensory elements of the dynamic sculpture can enhance awareness and promote discussion on climate change and sustainable production. This exploration raises the research questions:

- How should art position itself in the climate debate, considering the responsibility of material sustainability within the artworld?
- How can we create a dynamic sculpture that evokes emotional responses to climate change and through its aesthetics can contribute to discourse on environmental sustainability within the artworld?

For relevant literature, we worked with systematic literature searches, using carefully chosen terms, relating to the themes with various search techniques in different databases and online libraries. This literature was used throughout the project, for deciding on the literature.

The report on the project consists of a state-of-the-art section showcasing key inspirations for the project, one of which is *Ice Watch* by Olafur Eliasson, which has a unique possibility to start conversations on climate change and sustainability, and art's role in sustainability. The project uses the theoretical frameworks of John Dewey and Christian Jantzen, to create the design method and a basis of the analysis and discussion. Furthermore, will a section of the implementation and development of the sculpture be provided, an analysis of walk along interviews of exhibition goes and a discussion on the effects of *Beneath The Weight* brought into context of the discourse on art and sustainability.

2. State of the art

For our project, we have been investigating other people's work to find inspiration and ideas for what the project should be about. This section will examine art and sustainability, and how art can become a medium to raise awareness of certain issues regarding climate change and sustainability. Furthermore, we will explore some artists who have each inspired a conversation about art in sustainability, and how they have impacted society while also raising questions about their own approaches to this. Another part of our project is researching the connection we, as humans, have to sustainable issues and how we react to

news about this. This also becomes part of a necessary discussion on art's role in sustainability and how we have tackled it in our project.

Climate change is without a doubt a very present and pressing issue that we hear about in our world today. According to Keunhye Lee, this matter does not just belong to different types of scientists. It is a matter where art should play a big role as well. Contemporary art should make an effort to present these issues, as art influences and provides an understanding of the climate crisis on a tangible and engaging level (Lee, 2021, s. 1). Keunhye Lee intends to showcase and explain artists who make climate art as part of their practice. This is to highlight that there are artists in the field, but they should be much more present when we speak about environmental issues on a societal level, as they can have a very significant influence on people's understanding and reactions to this. This is noted by Lee in this quote: "This means that art does not provide a solution but can create a personal connection to climate change that can stimulate individual perceptions and awareness"(Lee, 2021, s. 1). Although Keunhye Lee's paper discusses art in public spaces, we found that his perspective could also apply to our project because he discusses certain things we are interested in as well. We have been testing our artwork under different circumstances, not in public, and therefore the focus has been a bit different, but we want to use this as part of a discussion on why it is important to display art and sustainability.

The whole topic of climate change has been rising in the art world. Many artists are creating experiences/artworks either based on certain issues or for specific causes. This also leads to artists finding inspiration and embracing more aspects of these issues by making tangible artefacts that are not just based on creativity, but also on real data and ideas from scientists, philosophers, technology, and the community. Artists use their work to depict and reflect on different aspects of our environment across time—past, present, and future. They do this through diverse artistic techniques, mediums, and frameworks. Through using art as a medium, "artists present our past, present, and future environment, using various methods, forms, and contexts, such as considering how humans impact our environment and how changing human behaviour could change our future"(Lee, 2021, s. 3). This report will further discuss the role of the artists and their artwork and how that could influence the way we talk about sustainability.

This leads to the first artwork that inspired our project.

2.1 *Ice Watch* by Olafur Eliasson

According to Olafur himself this is what he explains about *Ice Watch*:

These twelve immense blocks of ice are free-floating icebergs that have melted off from the Greenland ice sheet. Relocated here to the Place Du Panthéon and arranged in a clock formation, they present a tangible and immediate testimony of the dramatic effects on climate change (*Ice Watch • Artwork • Studio Olafur Eliasson*, u.å.)

Ice Watch has not only been displayed at Place Du Panthéon in Paris but has also been shown in Copenhagen and London. This piece of art has raised several dilemmas and even a paradox that we wanted to use in our discussion. In Lee's article, he fails to recognize that this piece of art was transported from both the Arctic and Greenland, which had a carbon footprint that is not being discussed. When it was shown in London, the ice blocks came from Greenland and were stored in nine refrigerated trucks and then driven to the installation site. Olafur Eliasson notes that he calculated the carbon footprint with a non-profit organisation to find the best solution for more sustainable transportation. Eliasson has defended the project despite the melting ice by comparing the carbon emissions of transporting the ice blocks to the emissions generated by a single person's round-trip flight to the Arctic to witness the melting ice. They aimed to highlight the environmental impact of individual air travel, especially in the context of mass tourism (Artverlaine, 2021, p.3).

This is also why we find this artwork interesting: it is time-based because it shows the rising temperatures in real-time, which in his own words is a tangible way to communicate these issues. It might be on a smaller scale, but it still has an impact on the environment.

Ultimately, the question of the artist's integrity in this context hinges on the balance between the immediate environmental cost and the potential long-term benefits of raising awareness about climate change. Eliasson's transparency and efforts to mitigate the impact suggest a level of integrity, but the contradiction inherent in the project's logistics remains a valid point of critique. This duality invites further discussion and reflection on the role of art in environmental activism and the responsibilities of artists in promoting sustainability.

It creates publicity and social engagement, which might be the intended idea of the artwork, but being part of leaving a carbon footprint also opposes the idea of sustainable art. This is where the artwork creates a paradox because the issues it is trying to communicate might not be recognized the same way if he did not show the actual ice blocks. In Costina Mocanu's article, she acknowledges that Eliasson is bringing glacier ice to get a public reaction, and he argues that it speaks to a large issue; therefore, one could argue that making this artwork is a form of investment in public awareness of a very significant issue. *Ice Watch* is then a more

intellectual work because it operates on multiple levels. There is social stimulation because you can interact and take part in the action, but you can also just become a spectator and watch it melt in real-time. Olafur creates a very real narrative which impacts us globally. *Ice Watch* becomes an analogy for issues far beyond our grasp and reality.

Another artist from whom we have found inspiration when speaking of sustainable art is Anish Kapoor and his exhibition at Arken. The exhibition is called *UNSEEN* and is running from the 11th of April 2024 until October 2024. Director of Arken Marie Nipper says in an interview:

Vi er meget stolte af, at Anish Kapoor har valgt at indgå i et tæt samarbejde med ARKEN om at præsentere nogle af hans mest imponerende – og teknisk udfordrende – værker her på museet. Enkelte steder har vi måttet bygge ARKEN om eller ligefrem bryde gulvet op for at få plads til de omfattende værker.” (*Spektakulær udstilling med Anish Kapoor på ARKEN bliver Skandinaviens største | ARKEN Museum for Samtidskunst*, u.å.)

It is worth noting that in this interview she explains how they had to rebuild the museum to fit the artworks and even dig into the floor to have the proper space required for the comprehensive artwork. This shows great respect for an artist like him, and his artistry does not compromise on any material scale. However, this also presents a challenge to sustainability. The need to balance artistic expression with environmental responsibility is a critical issue for contemporary artists and institutions. Kapoor’s work exemplifies this dilemma, prompting a discussion on how artists can innovate within sustainable practices.

In summary, when museums rebuild their spaces to fit installations, it reflects their dedication to artistic excellence, innovation, and enhancing visitor experiences. While this ambitious approach is necessary for showcasing contemporary art, it must be balanced with a commitment to sustainable practices. This is a topic we will discuss further in our report when we present our creative process.

We raise the question of whether artists need to take more responsibility to make art sustainable, or if we can compromise in other ways while still achieving the emotional reactions that can lead to potential change. Both Eliasson and Kapoor, through their ambitious and provocative works, challenge us to consider the balance between the artistic message and its environmental footprint. This compels the audience to reflect on the sustainability of art itself and the role of artists in highlighting and addressing ecological concerns.

3. Theoretical frameworks

This section establishes the theoretical foundation of the report. It begins with an examination of dynamic sculpture, followed by an exploration of John Dewey's concepts of art and experience. Furthermore, it delves into the theoretical framework provided by Jantzen, offering a detailed analysis of his definitions of experiences and their understanding through a psychological lens.

3.1 Dynamic sculpture

The dynamic sculpture art arose from kinetic art post war from the 1950's. Within this field multiple artists like Jean Tinguely, Bruno Munari, K.G. Pontus Hulten was a part of shaping this newer art field at the time. In Stephanie Jennings Hannor dissertation she explains the artists working in the mechanical art field describes their work as dynamic sculptures, this is defined as: “...defines machine art as encompassing works of art that move mechanically, in a determined manner, either by direct human propulsion or through the intermediary of a man-made machine and that incorporate or draw upon an industrial aesthetic.” (Jean Tinguely, u.å. p. 2) This led us to a discussion on how we can incorporate theoretical background on dynamic art for our project. Moreover, this also gave us the opportunity for us to use it in our artistic design process.

Jean Tinguely played with the notion of mechanical art and used it as his representation of how he saw the technological developments at his time and if there were any limitations he saw placed on the artworld. What is being noticed about Jean Tinguely's sculptures is that they were *useless machines*. This terminology was already being used by Bruno Munari in the 1930's, which simply translates to machines that have no practical purpose but instead emphasise the artistic and theatrical aspects (Jean Tinguely, u.å. p.3)

Tinguely utilises the concept 'mechanical performers.' This means that if one were to think about machines as mechanical performers, we would gain knowledge on how the piece of machinery works in a dramatic context. If we were to view machines as actors or performers, then they carry a certain role and function in settings that they are not typically found in. This is being highlighted by Hannor in her dissertation (Jean Tinguely, u.å. p.3) It makes Tinguely a relevant artist to reference as we are also making a mechanical performance that is supposed to have a certain role and function for the project and is placed in an atypical situation.

“Associated with the performative aspects of Tinguely's mechanical creations is an artistic fascination with the human-machine analogy in art” (Jean Tinguely, u.å. p.4) This suggests that there is more to the sculpture than being a performance. It reflects upon an artistic interest in comparing and connecting humans and machines. In our case we are making a

human-like sculpture with performance elements that should connect the viewer to, in our case, climate change, art and sustainability.

Additionally, Tinguely was not hiding anything in his works. They were displayed literally with the materials used as they were and natural sounds from the materials as well. This clearly indicates an honest work that doesn't try to be anything it is not. In our project as well, we want to be honest with our materials and we do not try to hide it. We will address this further in section 5.1

3.2 Art As Experience by John Dewey

This section provides an overview of the framework and principles from John Dewey's *Art As Experience*. These elements will be used to frame the design method and provide part of the theoretical perspective in the analysis of the experience with *Beneath The Weight* and in the discussion on art and climate.

According to John Dewey, one of the most fundamental parts of artworks, is that they are to be understood in relation to life. Dewey is writing about the importance of bringing the division between art and the mundane down, as this separation reduces their significance (*John Dewey - Art as Experience-Perigee Trade (1980).pdf*, u.å. p.3). Fine arts have been put on a pedestal which has separated the arts from the common or community. This is explained through the term aesthetic experiences, Dewey views this at its core to be an interaction between the environment and an individual, through an experience that places engagement in understanding, transforming the perception of objects and actions into a meaningful experience.

Dewey argues that "In order to understand the esthetic in its ultimate and approved forms, one must begin with it in the raw; in the events and scenes that hold the attentive eye and ear of man" (*John Dewey - Art as Experience-Perigee Trade (1980).pdf*, u.å. p.4), thus one cannot separate their immediate perception from the understanding. A spectator should not remain a cold spectator just seeing colours, but "should be carried forward, not merely or chiefly by the mechanical impulse of curiosity, not by a restless desire to arrive at the final solution, but by the pleasurable activity of the journey itself." (*John Dewey - Art as Experience-Perigee Trade (1980).pdf*, u.å. p.5). Thus, art must be brought into the common or community life, understood in this relation not merely an object with no connection. Creating an artwork for an event for the common life, thus provides the artwork significance, which can reach its fullest potential.

The connection between life and art is thus a unity, which creates an experience. Throughout chapter three Dewey works with the idea of having an experience. Within experience lies three elements, sensory, emotional and intellectual, which he delves into throughout the chapter. These three are the main categories of aesthetic quality. Sensory elements include sensing, taste, touch, seeing, smelling etc. This means it involves the colours, textures and

more, to enrich the perceptual encounter with art. The emotional element works with the emotional responses evoked, this is an important part of engagement with art, the perception and contributing to the significance of the artwork. The intellectual element works with the cognitive and critical aspects involved in experiences, this means how artworks stimulate intellectual reflections, interpretations and thoughts. These aesthetic qualities create an aesthetic experience with more depth and coherence, when they are intertwined.

Dewey speaks of two aspects of experiencing *doing* and *undergoing*, these works as a joint unity, with alternating between doing and undergoing. The two refers in its simplest explanation, of *doing* as an action which leads to *undergoing*, which in turn leads to properties of during. An example Dewey provides consists of the action of lifting a stone, which consequently leads to feeling weight and texture, which leads to a new action of what can be done with the stone. (*John Dewey - Art as Experience-Perigee Trade (1980).pdf*, u.å.p. 44) These works with active and reactive aspects of experiencing, working as a framework for understanding engagement with the world and art.

Working with sustainability and climate change, is working with the unification between art and life, as the sustainability subject is a common part of how we are to live our lives and is part of the forever important matter of our existence, future. Dewey writes “art breaks through barriers that divide human beings, which are impermeable in ordinary association”(John Dewey - *Art as Experience-Perigee Trade (1980).pdf*, u.å. p.244), which at its very strong wording, shows a clear indication to the idea that art can provide opportunities which other fields may not be able to. Sustainability can be difficult to talk about as much of the information is from political and economic fields, which at times may lack an emotional connection. Through the unity of intellectual, emotional and sensory components of aesthetic experiences, there is a strong opportunity of breaking barriers.

Dewey writes about *The Expressive Object* in the fifth chapter of *Art as Experience*, he indicates that “Expression, like construction, signifies both action and its result.”(*John Dewey - Art as Experience-Perigee Trade (1980).pdf*, u.å.p.82) What he means by this and is the point of this chapter of his book, is that true artistic expression cannot be understood by isolating the object from the creative process that created it, moreover expression also lies in utilising the materials as medium giving them new life and meaning. This creates a unified approach towards understanding how to work with a dynamic sculpture, and how to add meaning to all parts of the sculpture. Further this chapter examines the relationship between artists intention, the artwork, and the audience interpretation. This is done through a dynamic sculpture that communicates the emotional and intellectual aspects of climate change.

3.3 “Experiences and experiencing: a psychological framework”

Another theorist who speaks about experience is Christian Jantzen. We are focusing on the chapter “Experiences and experiencing: a psychological framework”. This text works as a continuation on Dewey’s theory, with a more in depth look on experiences, done through a psychological perspective. Here we will provide an overview of the principles that guided our research, thereby laying the groundwork for our methodological approach and how this can be implemented in the analysis and discussion.

Christian Jantzen explains “*Experiences are mental and corporeal phenomena.*”(Jantzen.pdf, u.å. p.146) Experiences encompass both psychology and physical aspects. Jantzen’s framework is founded on two assumptions “*Firstly, experiencing is a coherent structure of present sensing and feeling informed by past experiences. Secondly, experiencing is a specific structure by making us aware of the act of sensing and feeling and by questioning the validity of past experiences.*”(Jantzen.pdf, u.å. p.146) This introduces the concept of experience which is an essential part of our project. What is suggested in the first part of the quote is that experiences are not isolated or random, but based on previous experiences and suggests that our current sensations and feelings are shaped by our memories and previous encounters. The sensations and emotions we experience at any given moment form a unified whole and give us new experiences and memories. Furthermore, this can be used when talking about the immediate sensations and emotions a visitor experiences when they are interacting with, in our case, the dynamic sculpture. The dynamic sculpture should be designed so it can fully engage the visitors’ auditory senses and evoke emotions. The second part of the quote highlights that experiences are not only coherent, but also distinct in a way that they draw our attention to the sensory and emotional process themselves and makes us question the past experiences and if they are still valid. When we become aware of the act of sensing and feeling, we start to pay closer attention to how we are experiencing the world around us. If the work creates awareness it can lead to new insights and deeper reflections. This would potentially enrich the audience’s experience and add a layer of cognitive engagement that could make the experience more memorable. This theoretical framework can be effectively applied to our design of a dynamic sound sculpture to create a richer, engaging, and reflective sensing and auditory experience.

When designing a framework based on Christian Jantzen concepts of experience it is important to note that he uses different definitions of experience depending on the situation. He explains the different definitions of the concept of experiencing by referencing the Merriam-Webster dictionary. We have found that definition 1 and 5, are the most relevant for our project. The definitions are as follows:

“1. ‘direct observation of or participation in events as a basis of knowledge’ (Jantzen.pdf, u.å.p. 150). This definition encompasses experience as evidence, this means that our direct encounters and observations help us determine the truth of propositions or facts. In addition, the truth of any statement given by participants are verified through their personal experiences and empirical data that states this as truth. If we experience something first hand, we are inclined to consider it true until someone else proves us wrong with other facts. This concept of experience is supported by our sound design as it states facts and data about the condition of our climate change situation. This means that the participants doing our walk along would be inclined to acquire knowledge about the climate situation and use their own experiences as well to verify the experience.

“5. ‘the act or process of directly perceiving events or reality’.” (Jantzen.pdf, u.å., s. 150) In other words, this definition differs from the first definition. Here the focus is on solely the immediate and subjective experience. Additionally, Jantzen also roots in Wilhelm Wundt’s concept of “immediate experience”, which focuses on our personal, subjective feelings and perceptions in each situation. Opposed to the first definition, the evidence here is subjective and comes from within us. It is about our internal reactions and presence in an event rather than external observations. We find truth from our internal and personal involvement and emotions in the situation, despite whether it can be “falsified” or objectively measured. This will also be supported in our walk along as we invite participants to reflect on their own subjective and immediate experience of the artwork.

Table 8.1 Comparison between the English ‘experience’ and its German equivalents

‘Experience’	German equivalents
Experience (1): Knowledge through observation	<i>Erfahrung</i>
Experience (2): Practical skills	
Experience (3): Conscious knowledge	
Experience (4): Autobiographical knowledge	
Experience (5): The act of sensing and feeling	
	<i>Erlebnis</i>

Figure 1 Experience and the German equivalents by Jantzen

Table 8.2 Erleben, Erfahren, Erlebnins and Erfahrung compared to the five meanings of the noun 'experience' identified by Merriam-Webster and to categories of memory in cognitive psychology

'Experience'	German equivalents	Categories of memory
Experience (1): Knowledge through observation	<i>Erfahren</i>	Implicit procedural memory
Experience (2): Practical skills	<i>Erfahrung</i>	
Experience (3): Conscious knowledge	<i>Erfahrung</i>	Generalized episodic memory
Experience (4): Autobiographical knowledge	<i>Erlebnis</i>	Autobiographical episodic memory
Experience (5): The act of sensing and feeling	<i>Erleben</i>	

Figure 2 Extended Experience and the German equivalents by Jantzen

Jantzen discusses how merging experience 1 and 5 has its advantages. (Jantzen.pdf, u.å., s. 151) First of all he explains how epiphanies combine both objective (1) and subjective (5) aspects of experience. Participants experience both factual evidence but are inclined to form their own opinion as well. Merging these experiences (1) and (5), participants are acquired with a more coherent experience.

Christian Jantzen points out that there are German equivalents to broadening the understanding of what experience is. He presents in figure 1(Jantzen.pdf, u.å., s. 151) two German equivalents that fits experience *Erfahrung*(1,2,3), *Erlebnis* (4,5). He specifies by inflecting experience and compares how they match with the Merriam-Webster dictionary, this is shown in figure 2(Jantzen.pdf, u.å., s. 155). What is equivalent to experience 1 is *Erfahren*, and the equivalent to experience 5 is *Erleben*. *Erfahren* (1) refers to accumulated experiences or learning over time. It also speaks to creating value with factual remedies for the participants to bring awareness of the situation and potentially lead them to a discovery or experience that would make them reflect upon what they gain from experiencing this and potentially make changes.

Erleben (5) signifies the act of experiencing in the present moment, involving direct sensory and emotional engagement. It is the process of sensing and feeling in real-time. This focuses on the act of directly perceiving events and reality. The objective experience you gain from *erfahren* (1) enhances the experience (5) of sensory and perceptual connection between participants and the themes of sustainability.

In the next section of the report, we want to describe how we will use the notion of experience through Jantzen and Dewey and how this can be used in the design process, as well as a framework for the analysis and discussion.

4. Design method

4.1 The notion of experience

This section will describe the design method which will be used for developing the dynamic artefact. The method is using theoretical perspectives of Dewey and Jantzen to create an emotion evoking and an inspirational work about climate. The method is working with insights particularly on the perspectives of the aesthetic experience, the connection between common life and environment, in an experience promoting reflections for the audiences. It will also work with the sensing and feeling, and the perception described as experience (1) and experience (5) respectively in the previous section. This will inform the design principles and will be used to ensure that the artefact is engaging the audience with climate.

Based on the theoretical background we have certain goals of the project. We want, through the notion of experience (1) and (5), to achieve both awareness on climate change but also evoking emotional responses to the climate art debate, thereby also inspire to focus more on a sustainable art production in the artworld. Additionally, we want to create a framework for how we designed our artwork. Furthermore, we want to integrate John Dewey's principles of "art as experience", and Jantzens *Erfahren* and *Erleben*.

The first part of the design phase was gathering the accumulated experiences (*Erfahren*), here we would discuss and engage with factual information and past experiences regarding sustainability. The research drawn from this phase would become part of reflecting on what to do for the next phase. In the next phase we were brainstorming on how we could create an immediate sensory engagement (*Erleben*). This was to ensure that we could develop an immersive experience that would allow participants to engage their senses and emotions. This phase can help the artists find a solution on how to present a dynamic experience for the audience. In this phase it was crucial to experiment and test as we also had to find a connection between human and sustainability. During the forthcoming phase we would add elements like sound to test how we could achieve certain experiences (5) for exhibition goers. Here we also benefited from having a collaboration with fellow students who could help us in testing our iterations. The very last stage is meant for feedback, which would be gained from a walk along with participants. The feedback session makes it possible for us to improve and enhance our artwork. Next, we want to describe the walk along and how we conducted it.

4.2 Walk along

A method evolving evaluating people's experiences with artworks is a walk along. This method has been constructed by Bikuben. This method works with understanding people's experiences of art and thoughts on an artwork or exhibition. The purpose of a walk along is

getting insights on what happens when participants experience art and what reflections they have while experiencing the artwork. To conduct the method, it is necessary to create a relaxed and comfortable atmosphere, this is to ensure that people who have agreed to do this are not feeling forced into certain emotions or feelings when experiencing. This method is not exclusively for people with a strong knowledge of art, the method ensures that all types of exhibition goers are able to participate in talking and express what they experience when seeing an artwork. This was relevant to our exhibition as well.

According to Bikuben this method can be used to go through an entire exhibition but as it is up to the individual participant of what they want to talk about, it can be used for just a part of the whole exhibition. As one of the main points of the walk along for this project is the conversation between art and environmental sustainability, having the participants walk around to any exhibited artwork, would lead the main objective out. Therefore, we invited them to talk about the artwork, considering it as a solo exhibition. From these participants we invited them to move closer to the artwork and let them move around as they pleased within the exhibited area.

As previously stated, this method builds on making the conversation part comfortable, which we also adapted for our walk along. We let the participants decide how we were going to conduct the conversation by allowing them to sit down or walk around. We started the conversation by asking lighter questions advised from Bikuben. They suggest using “Hvad minder billedet dig om?” (*Bikuben_Kunstens betyding metoderedskab.pdf*, u.å. p.12) Bikuben advised further that it would make sense to present the point of the walk along in the very beginning and let the participant decide how long the walk along should be. In our analysis and discussion, we will investigate the responses from the participants and how we can use it to enhance the project even more or even discover new things.

5. Implementation

Following section will describe the implementation, including an overview of the starting point, to the development and finalising of the concept, and the execution of the prototype, including the building of the sculpture, hardware and programming, and lastly the creation of the sound.

5.1 Development

We had two overall approaches to sustainability and art. We had a perspective of public engagement, as it was created for a festival about climate, thus art as a medium for sustainability conversations. Moreover, we also had a perspective of sustainable art production. We wanted to focus on the first perspective for the concept, where the second is a subsidiary focus, used for the discussion section and dynamic sculpture's design choices. These approaches came to light early in the process as we through a collaboration with

Aalborg Kommune, Nordværk and discussions with another group of students, who are also collaborating with the two aforementioned, and with a theme on art and sustainability.

Right from the beginning of the project, before final groups were even made, we decided to create an artefact for the climate event *Aalborg2024*, this provided constraints, including size, durability and other physical aspects, and concept constraints as it had to fit in with the frame. These constraints were used from the beginning in brainstorming. Having ideas of environmental sustainability of the project led us to look at what materials already existed. Through this we looked at artworks which worked with sustainability or climate, one of the artworks we found was *Ice Watch*, we found it interesting with the concept of moving ice blocks from Greenland to Europe. This showed us the idea of how art influences sustainability, both with how art can be used in the conversation of sustainability and how art can contribute to climate conversation in a negative view, for instance the transportation or production. After having the opportunity to work with the local recycling station from our collaboration with Nordværk, we decided to work with the premise of the materials being sustainable. After choosing that the materials for the final product should be a big part of the project, aesthetic constraints arose to fit within this material sustainability. As we had decided on using materials from Nordværk, we went to the recycling station to look at the different materials, with a mindset of aesthetic possibilities but also practical solutions. Notably, we were not entirely sure of what was possible to get from the recycling station as well.

Having decided that the materials being a determinant of the physical appearance, also led us to become very mindful of this theme, while creating the artworks. Large parts of the creation were tedious work, which gave us a lot of time to also focus on the idea of the project. From this we became more self-aware of the artistic practice within art. It led us to think further of our education and how our art production in previous semesters haven't had a focus on how art plays a role in sustainability. Previously, artefacts working with sustainability worked with an educational/informative focus only, while not really taking the actual production into account. Hence why we decided to make this one of our primary decisions for the design process. The decision on sound also made us aware of the information of climate and sustainability we are experiencing now and have heard of. We had conversations about how we felt that there was information on climate everywhere we went. This result led us to taking a walk in Aalborg, looking at what information on sustainability we would encounter. Most, if not all our encounters, were from Aalborg Kommune, especially on how to separate waste products. We saw no other information, except for these action initiatives. This just further acknowledged our feeling that most of the information we encounter has an action driven component trying to nudge people to do something for the climate. This overwhelming amount of information feels like a toll on the body. In this way our own experiences with everyday occurrences fostered a strong connection to creating an artefact and giving opportunity to have an experience on some of these awarenesses we met, through the process of developing the artefact. This also supports what Jantzen says about experience (*Erfahren*), it is something that is accumulated through previous experiences which was our walk around in Aalborg and how it led to fostering a new connection towards art and sustainability.

The concept of the artwork thus resulted in working with this emotional toll. We were inspired by the artwork *End of Tears* by Yui Ishibashi, where a person is sitting overwhelmed, consumed by nature. This sculpture signifies the emotional toll and was shown through a physical reaction, which we also saw ourselves, as we were doing a mini workshop between ourselves, where we worked on understanding our own physical and emotional reactions to the climate, here we realised that we both had a physical movement of crouching down.

We were inspired by Jean Tinguely's dynamic sculpture, to the material honesty, where all materials are visible, together with the notion of *useless machine*, where the movement has an artistic rather than practical character. These elements led us to a concept working with the emotional toll, being put into physically moving objects, which would be overwhelmed by information. See figure 3



Figure 3 Sketch of concept

5.2 Execution

5.2.1 Physical Artefact

The Artefact has been made from mostly recycled material, it is made in a life size of a human being with the appearance made from visible wires and Styrofoam. (figure 4, 1) This has been with the intention of being very open and direct in the use of materials as well as speaking on the production of the artefact.

The artefact has movement which we had to consider in the making of the artefact (figure 4, 2). Because of this the movement with the motors were created first, where to the appearance were added later. The body was created through Styrofoam cut and shaped based on accrual measurements, to ensure a realistic physical appearance (figure 4, 3). We didn't want to create a one-to-one person but rather have a human like artefact, which is very clear as the appearance of the artefact does not draw strictly on realistic human appearances. To decorate the physical appearance we have used wires, looking at the possibilities of appearance we were stroked by the colour possibilities in wires (figure 4, 4). These wires became the appearances for the face and legs (figure 4, 5). For the hands and feet, we let the Styrofoam be exposed, leaving a bit of material honesty, inspired by Tinguely, with the intention of people being able to see the materials in the work. The choice of having a shirt covering the torso, was partly due to the torso having to be able to move, and having the arms move along



Figure 4 top left corner(1): scale of sculpture next to a person. Top left corner(2): skeleton for moving element of the sculpture. Bottom left corner(3): working progress of shape the sculpture. Bottom Middle(4): Experimentation of colourful wires. Bottom Right corner(5): design of face with blue wires.

with it. This meant using a material, which could have fluid movements were practical, it also enabled us to have easy accessibility to the electronics hidden in the lower back of the sculpture.

5.2.2 The hardware and programming

The electronic components hidden in the back enabled the body to move up and down in accordance to the soundscape played on speakers. To create this, a linear actuator has been used for the mechanism for raising and lowering the body. To create a likeness of body movement, comparisons to a real body have been made, in terms of the degrees in which the body could move. This actuator has been connected to an H-bridge and an M5stack, also attached to the M5stack is the sensor. See figure 5. The M5stack is connected to a router placed hidden in the exhibition room together with the computer, which is running the Processing sketch, and playing the audio through the speakers.

The programming of this work consists of two sketches one in Arduino and one in Processing. The one in the Arduino is controlling the motor and the one in processing is controlling the sound.

The Processing sketch analyses the "loudness" of the looping audio file of the sound in the dynamic sculpture. It is visualising the "loudness" though a size changing circle (for debugging purposes) and sends the data of the "loudness" to a remote location via OSC (See Appendix 1.) For the processing sketch the sound analysis is made from the build in example *PeakAmplitude* from the sound library. In addition to this an oscP5 library is used to send the information to the M5stack. The Arduino sketch connects to a wifi network and subscribes the OSC messages from the Processing sketch to receive target angles. It uses a BNO055

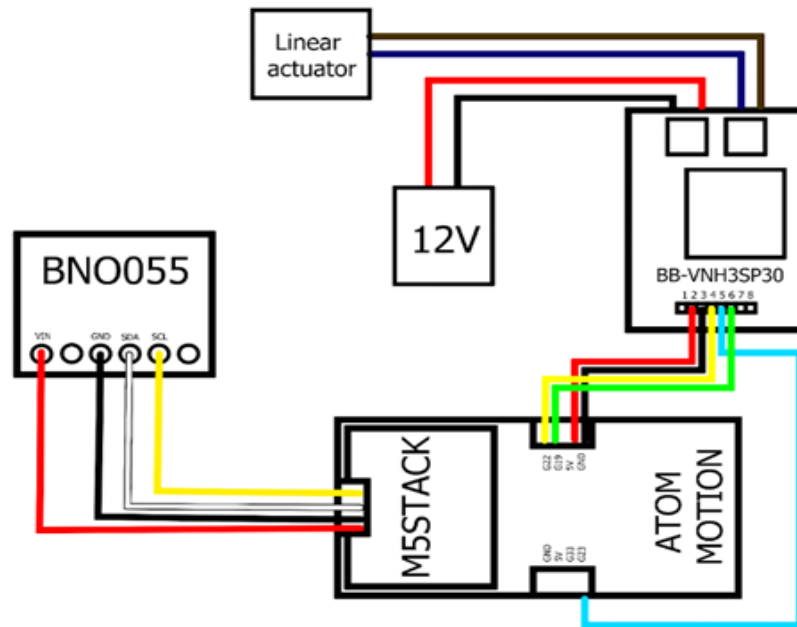


Figure 5 Overview of electrical components

sensor to read the angle of the “back” and adjusts the motor's position, so it moves to the correct position of the OSC message. The motor's direction and speed are adjusted to minimise the error between the current angle and the target angle. The system updates continuously and prints relevant data for monitoring, see appendix 2. To control the smoothening of the movement of the motor the P in PID (proportional–integral–derivative controller) is implemented into the code. This is applied to minimise the error proportionally to the differences between the desired angle and the current angle. This means if the current angle is close to the desired angle, it will move slowly and if the current angle is far from the desired angle, it will move fast.

The ‘err’ is scaled, ensuring it is a positive number and constrained within a range suitable for controlling the motor speed.

```
int p = constrain(abs(err * 80), 0, 255); // Calculate motor speed (PWM) based on error
analogWrite(19, p); //sets motor speed
```

These two sketches made up the programming to make the sculpture move with the sound, which creation will be explained in the next section.

5.3 Sound

The sound design was created with multiple layers of different audio files and had a duration of 3 minutes. Since the motor was going to move with the sound, we wanted to create a sound design that developed and got more and more intense, which then led to the motor moving.

The sound consists of voices speaking about climate, and non-diegetic sound, consisting mostly of nature. All the conversation is created from news. We selected the news by finding articles and information videos from the Danish Broadcasting Service (Danmarks Radio), The Guardian, BBC and other channels. None of the news are more than a year old, as we wanted to keep them as close to what we are experiencing now. To play with the uncanny nature of our sculpture we decided to use AI voices to experiment. The structure of the sound was created with a mindset of the amplitude, which was used in Processing to control the motor. The structure for the sound was created with a rising intensity until the end of the sound to then fade down. This structure resulted in the movement of moving up and down in a pattern, when being analysed in Processing.

We combined amplitude structure with the content of the sound, which resulted in six phases. Five phases had an increase in intensity, both with the amplitude and the words. The last phase declined in amplitude and news intensity, ending the duration with a positive note. See Figure 6 for visualisation of amplitude and phases for the structure. The sound thereby had a narrative structure of the news but is not a narrative storytelling. The following passage explains each of the phases in more detail.

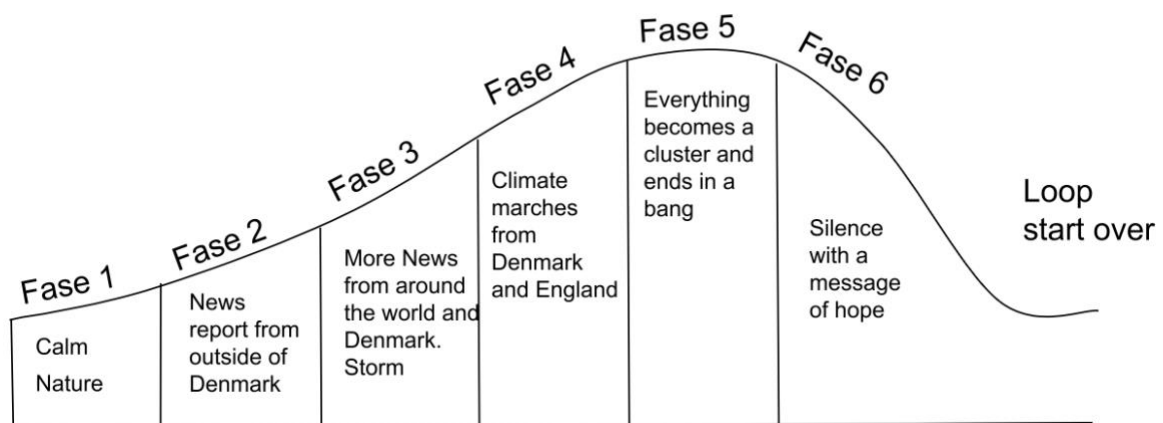


Figure 6 a model of how the sound design was constructed

The first phase is calm and represents the idyllic, but also realistic nature we hear around us. This was inspired by the surroundings we have in Aalborg.

The second phase added a singular news that told a rather big issue happening somewhere else in the world. This was inspired by the butterfly effect. A butterfly flaps its

wings in one place and causes a tsunami on another side of the world. This was to emphasise that we are so used to hearing climate issues from far places in the world, so far away from our daily life.

The third phase adds more news both about issues in Denmark but also issues around the world. What is worth noting is that we found positive news about climate change as well because we thought it was necessary to also shine some light on some positive development. Though the sound gets cluttered, and it is difficult to hear the good from bad which also emphasises all the different types of news we get, but also what the press decides to focus on.

In the fourth phase there is another layer added with two different climate march demonstrations. One in London and another in Copenhagen. This was to show the multifaceted type of news, bringing attention to an issue like this. We wanted to show different points of view to allow recognition of both local and global issues, while making sure that we had different aspects and layers to news.

In the fifth phase, the last part of the buildup, we encounter even more layers that are experimented with. There are unnatural sounds, and everything gets more and more cluttered. The idea was that in the end it would be nearly impossible to understand what was going on and what it said.

The final phase makes us listen to a guy reporting that good things could happen in the future if we, with each other, support and research how we can do better for the environment. It also leaves a little hope for the future. It was really important that the sound did not become too dark because there is hope for the future.

The combination of all beforementioned parts created the prototype of the dynamic sculpture *Beneath The Weight*. This was then exhibited at Create a campus of AAU, where we conducted walk along interviews with participants to analyse the prototype in relation to the discourse on art and sustainability. Following section will provide more on this.

6. Analysis

We want to provide a comprehensive analysis that focuses on delving into both our observations and the responses given by our participants from the walk-along. We will ground our analysis in the design method described earlier, with a focus on the sensory, emotional and intellectual perspectives informed by the theoretical perspectives of Dewey and Jantzen.

The walk along interviews were conducted according to certain criteria regarding the participants. Given the problem formulation's focus on sustainability and art, it was relevant to engage with individuals with varying levels of interest in these areas. Thus, we were

selecting participants, with varying degrees of art interest, from rarely occupying themselves with art to working with it daily.

Before the walk along each participant had the opportunity of exploring and viewing the artwork. The walk-along commenced at a slight distance from the sculpture, allowing participants to move towards the area they found interesting. During this process, we engaged in conversation, guiding them towards topics of the aesthetics of the sculpture, sustainable production, and the intersection of art and sustainability. Each interview had different content and questions tailored to the participants' interests and themes they introduced.

To analyse the interviews, we reviewed the audio recordings and categorised the responses into various coding categories: descriptive coding, language coding, metaphors, symbols, and conceptual coding. We then aligned these responses with objectives from the design method. We will conduct a thorough qualitative analysis, based on these categories, to gain deeper insights. All participant answers that will be referred to can be seen in the appendix 3.

6.1 Sensory and emotional

One of the primary objectives of the design process was to elicit sensory and emotional responses from participants towards the dynamic sculpture. One of the first things that was pointed out was that all participants reacted to the human-like appearance. There were mixed reactions to seeing it initially, some described a shock of seeing a person, but then realising it is not a person, while others described the human-likeness with the uncanny valley concept *"Creepy og uncanny valley, man kommer hen og den er humanlike figur, og så bevæger den sig lidt."*

According to the Cambridge Dictionary, the uncanny valley is described as
the unpleasant feeling that some people have when they see robots (machines that can carry out actions automatically)... Uncanny describes a situation in which a machine looks so eerily 'almost but not quite human' that it just creeps people out.
(*Uncanny Valley*, 2024)

Thus, the participant felt an unpleasant feeling due to the almost human appearance. However, others did not perceive this uncanny effect, possibly because the wire structure made it clear that it was not human-like, despite its human proportions and posture. This aligns with Dewey's notion that an expressive object incorporates and communicates the artists' intended, as evidenced by participants' mixed reactions. This indicates that the decision on recycled materials and visible electronic components in the sculpture effectively conveys the environmental message intended by the artwork, thereby enhancing the expressive qualities of the materials used. This demonstrates how participants experienced more sensory and emotional impact.

The uncanny effect provoked feelings of the participants upon encountering the sculpture. However, based on the participants' movements during the walk along, none of the

participants showed signs of moving away; instead, they moved closer to examine it. This suggests a balance between the uncomfortable and intriguing aspects of the sculpture. During the walk along, some participants crouched down to see it at an eye-level, as Dewey posits it. The sculpture's human-likeness also led participants to quickly attribute human characteristics to it, as participants described the sculpture with gendered pronouns (he/she), personifying it despite knowing it was not real.

The sculpture evoked perceptions of it having actual feelings. One participant remarked, *"Man kan mærke hvor presset personen er der bukker sig"* suggesting that human feelings of stress and burden are being projected onto the sculpture. These projected human-like emotions led some participants to feel compassion, relating to the perceived suffering or struggle of the figure. Thus, the artists bring a total and unified experience, merging sensory and emotional responses with an effective environmental message.

6.3 The Auditory Elements

The primary focus of the sound is to allow people to experience the intensity of the pressure, rather than focusing on the actual words spoken. This is clearly reflected in the interviews, where participants consistently noted the sound, describing it as chaotic or overwhelming. One participant mentioned the absence of a narrative story with a message *"Lyden lyder kaotisk, lyder ikke noget sammenhængende, ikke nogen lineær historie med et budskab."* This indicates that the narration and words were not clear for the participant, suggesting that the spoken layer of the artwork was not comprehensible to all audience members.

When asked of the actual words spoken, some of the participants said that they either did not really pay much attention to them, only caught parts of it, or only listened after observing the sculpture for some time. Therefore, the actual words were less important to the participants in experiencing the work, with the emotional impact of the sound being the primary focus.

Participants observed how the sculpture moved in sync with the sound, feeling stressed on its behalf. They highlighted the figure on the floor succumbing to the pressure and intensity of the soundscape. This suggests participants found the work immersive, as they described feeling the pressure and intensity themselves. One participant articulated this by stating, *"Vægten man kan føle som individ i hele den her nogen gange uoverskuelige klimakamp"*.

Another participant described entering the room without initially noticing the intensity increase, but later noticing how loud it had become, stating *"Baggrundsstøjen, det bliver meget forstyrrende"*. Participants also noted that the loudness of the sound was when they observed the sculpture moving in sync with it. One participant remarked that the sound seemed to emanate from the sculpture itself, saying, *"Har det som om lydene kommer fra skulpturen."* This could partly be due to the correlation between movement and sounds, and the fact that the speakers were not visible to the audience members. Perhaps, this could reduce the immersion of sound, as the sound seemed to just be in the room rather than from a visible speaker.

The interaction between the sculpture and sound was created a more intense experience of the artwork, than either element would have alone. This holistic experience was expressed by one participant, who noted that the two elements together created a "*Helhedsoplevelse*".

6.4 Intellectual engagement

The conversations with the participants sparked themes and reflections, that were grounded in their personal knowledge and previous experiences. Jantzen discusses in *experience* (5) how previous experiences influence new experiences, creating an integral aspect of the discussions with the participants. For instance, one participant appreciated that the sculpture's colours derived from the materials themselves, avoiding the use of toxic paint, as they describe how heavy metals is a major pollutant, and one of the biggest sinners in the marine is paint, which they had a focus on because of their background. This added a reflected layer not apparent to all audience members. When discussing the materials and rationale behind their selection several participants noted that the wires served a purpose beyond mere appearance, contributing to the overall concept. One participant remarked, "*Lavede af genbrugsmaterialer: Når man får konceptet bag det giver det et godt formål*". This indicates that participants found the idea of creating environmentally friendly artworks compelling. Such discussions seem to facilitate conversations on sustainable production, as the artwork promotes sustainable practices. Which also reflects Dewey's idea that an expressive object should integrate sensory, emotional, and intellectual elements to gain a unified aesthetic experience.

A participant who works with art daily, described a reflected on the production and origin of material for their art projects, noting that typically do not consider these aspects in mind but recognised their relevance and potential influence on their future work.

Having participants with diverse views and interests of art generated different perspectives on the relationship between art and climate, and how they should influence each other. However, all participants shared similar overall responses to the aesthetic experience. Dewey emphasises the importance of harmoniously integrating the emotional, sensorial and intellectual elements of an experience. The experience of the artworks seems to fulfil these aims, as the different elements intertwine and prompt conversations among participants. Within the exhibition context, the artworks facilitate discussions that can extend into the broader community. The exhibition setting opens dialogues that can be further explored, like how "*Ice Watch*" has been used to initiate conversations within the art community and our group.

The analysis provided an in-depth understanding of the participants' perceptions of the artwork itself and its relation to a broader societal context.

7. Discussion

John Dewey emphasizes the importance of integrating art with everyday life, arguing that this integration strengthens the aesthetic experience. He asserts that when an artwork is isolated in a museum, it loses its full potential. We consider the climate debate and climate change to be integral parts of our daily lives, continuously influencing us through news, social media, and even our daily routines involving food, waste, and transportation. This theme profoundly impacts our day-to-day existence.

Many contemporary artists address various themes from their daily lives, with climate change being a prevalent topic. Numerous artists explore the theme of climate in diverse ways. The article by Lee, mentioned earlier, exemplifies how art not only incorporates everyday life but also how everyday life incorporates art. The article aims to study the impact of art on sustainability and the various ways art addresses sustainability. It explains how art is bringing conversation to the subject and creates spaces for conversation reflecting Dewey's idea that art should bridge the gap between everyday experiences and events, thereby enhancing relevance and impact.

One artwork that engages with the themes of sustainability, particularly transportation, and its impact on the climate is, *"Ice Watch"*. This artwork is controversial due to the questions it raises about the sustainability of transporting ice over long distances, only for the material to eventually disappear.

During discussions on art and sustainability, a participant questioned whether Eliasson could have created ice cubes locally and allowed people to believe they were from Greenland, focusing on the concept rather than the practical transportation. This could be argued as a more sustainable solution. However, the artwork's effectiveness might stem from this very controversy. The intense conversations and emotional impact it generated might not have been as strong if the artwork relied solely on a conceptual approach. Jantzen underscores the importance of creating an emotional connection to enhance the experience, and Dewey's perspective helps us understand that controversy becomes a significant part of the expressive quality. Thus, the artwork's success likely arises from the experience and reflection it provokes.

Our project, inspired by *"Ice Watch,"* led us to question the role of art in the climate change debate. We identified two general positions: one, that art should be created with mindful

material use, adhering to a cradle-to-grave concept; and two, like "Ice Watch," that art should generate critical conversations about pollution in art and other industries, prompting audiences to reflect on the ethics of transportation and similar issues. Given our collaboration with Nordværk and our inspiration from "Ice Watch," we were drawn to the first aspect—creating art without contributing further to pollution, or at least minimising production pollution. We purchased only a few materials, such as adhesive products, glue patrons and wood glue. We wanted to be transparent with our material choices, providing information at the exhibition about the arguments for the materials used, intending to be genuine and transparent about the origin and purpose of the artwork post-project.

These material choices heightened our awareness of our impact and influenced the entire project, leading to a continuous process where each aspect informed the next. As artists create their works, they critically evaluate and adjust their projects, ensuring coherence and connection throughout. Dewey notes that *"an incredible amount of observation and of the kind and of intelligence that is exercised in perception of qualitative relations characterizes the creative work in art"* (John Dewey - *Art as Experience-Perigee Trade* (1980).pdf, u.å., s. 50–51). The initial concept must be brought into the physical world through decisions that make the sculpture belong in the common world. "The real work of an artist is to build up an experience that is coherent in perception while moving with constant change in its development"(John Dewey - *Art as Experience-Perigee Trade* (1980).pdf, u.å., s. 51). The previously described decisions showcase the thought process and work involved in creating an experience. From participants' responses in the walk along, it is evident that they experienced what we had intended. We effectively translated our imaginative concept into the physical world. The entire creation process is an experience involving sensory, emotional, and intellectual engagement from the artist, as "the artist embodies in himself the attitude of the perceiver while he works" (John Dewey - *Art as Experience-Perigee Trade* (1980).pdf, u.å., s. 48).

Reflecting on the entire process of creating the sculpture, from start till finish therefore is an important experience itself. "Going over an experience in mind *after* its occurrence, we may find that one property rather than another was sufficiently dominant so that it characterises the experience as a whole."(John Dewey - *Art as Experience-Perigee Trade* (1980).pdf, u.å., s. 37). This quote explains that through the entire process of creating the artwork, we can look back on the experience with certain properties being more dominant than others. Those could for instance be the understanding of how we are affected and what feelings we have towards the climate discussion and how we are confronting it. Creating an artwork that is working with a topic relevant to most people today, we can draw art and everyday life, enhancing opportunities of appreciation of everyday life. As artists we play a crucial role in creating artworks that resonate with others, providing valuable experiences for both artists and audiences.

Using *walk-alongs* provides an opportunity to understand the audience's experience. Dewey's explanation of experience highlights the need that an experience should be a harmonious unity of sensory, emotional and intellectual engagement which we aimed to achieve through walk-alongs to understand the art-goers' experiences. Through these *walk-alongs*, participants actively pointed out features of the work and described their experiences, aligning with Dewey's concepts of doing and undergoing. Thus, working with walk-along interviews can provide meaningful insights into understanding experiences.

As the sculpture aims to provoke reflections and thoughts on climate change, its elements are carefully designed to work with these reflections. The soundscape has been layered in a way where the soundscape at first is relatively calm, and gradually becomes more and more intrusive until one cannot help but notice the sound. At the same time the movements of the sculpture can create a connection to the sound. Which audience members then are enticed to react and reflect on. The human attribute of the character enables audience members to empathise themselves with the sculpture. All three of the elements - sound, movement, and human likeness - are coming together to create a performance-like experience. These elements are intended to evoke emotional empathy towards the sculpture, through projecting emotions onto the sculpture. As Dewey states, "*In order to become emotional they must become parts of an inclusive and enduring situation that involves concern for the objects and their issues*"(John Dewey - *Art as Experience-Perigee Trade (1980).pdf*, u.å., s. 42). This setup compels the audience to emphasise with the sculpture, mentally placing themselves in a similar situation of feeling overwhelmed by sustainability, prompting reflections on their own stance on the issue.

As the dynamic sculpture addresses sustainability, incorporating scientific elements can strengthen the artwork's impact. "*Beneath The Weight* " can bridge the gap between art and science, as the artwork can communicate scientific knowledge on the environmental issues through an aesthetic experience. Dewey states, "*Science states meanings; art expresses them*"(John Dewey - *Art as Experience-Perigee Trade (1980).pdf*, u.å., s. 84). Thus, the collaboration of art and sciences can work within two different modes. As art expresses, it can address the statements from science in an accessible way. "*Beneath the Weight*" can express some of the scientific statements brought through the artwork's soundscape, through for instance, "*EU-rapport: Ekstremt vejr kostede EU-lande 100 milliarder sidste år 100 milliarder kroner og 151 menneskeliv. Så meget har oversvømmelser, storme og skovbrande kostet EU i 2023.*" This scientific element also promoted a sustainable mindset in our project.

One of the insights we gained from after our *walk-along* was within sustainable production. While we consciously chose colours based on the materials, we had available, we realised that our project did not add unnecessary products such as paint into the work, but rather used

the colours of the wires, and Styrofoam as a basis for the project. This is an interesting aspect of working with a sustainable production, which led us to work in a different way to what we might otherwise have done. The conversations with participants made us more aware of aspects of climate and art we had not previously considered, such as minimising the use of adhesives. One of the main points in the earlier process of our creation was to use the properties of the Styrofoam's pierce ability with the wires' ability to pierce. This proved to be affecting both the aesthetic look and the time-consuming labour, which ultimately led to us using glue. Although we didn't use a hundred percent recycled/lent materials, as the primary materials were recycled, the message of sustainable production was kept in mind as well. When thinking of art and sustainability, questions such as "should it be all or nothing?" in terms of how to approach the main aspects of what an artwork is working with. Based on what we saw from the *walk-alongs*, the concept of using recycled materials and using it in praxis was more relevant and interesting than the fact that not all aspects of the artwork was recycled. Thus, to convey the messages of the products artists may not necessarily need to be a hundred percent sustainable to speak of the production. Perhaps the decisions and transparency are enough to speak of the theme. Perhaps the fact that not all materials can be used as an engagement of thinking of solutions of using even more materials in a sustainable way, both within the single art piece but also perhaps the audiences' own lives.

8. Future work

This section will explore the life of our sculpture beyond our exhibition at Create, including decided plans and other possibilities for future work.

When we will exhibit at Kildeparken, we are able to use the conversations and reflections from the participants and other audience members for improvements of the sculpture. Some of what came from this included that if we could add more languages, we could reach more people, as the festival invites people from the EU. We could investigate making the experience much more intense by either adding more elements like lighting or having the soundscape speak more clearly about the sustainability issues.

As the core principles of the project are rather simple, it is possible to make changes to the artwork so that it could fit into a certain context of a specific exhibition or festival. This could be the aforementioned language, so the languages fit with the expected audience. Changes to sound could also prove relevant, as we could change the words to fit into a broader or smaller theme. Thus, it could work for an exhibition about climate change or wildfires caused by climate change. It could also be working with what words that were in focus at certain times to ensure more intense experience regarding a specific topic or event. We could even change the degree of provocative statements to get noticed even more. A notable example of giving our sculpture another life could be to re-purpose the theme/topic that it is trying to convey.

For instance, we could make the sculpture stay positive for a longer time until the world would crumble down. Perhaps working with the idea of living blissfully not having to be concerned with the effects of climate change, until one day when it comes close to you.

All these changes would be digital aspects of the sculpture, what would be needed to make them work, is to create a new sound file, which means that the sculpture could take different shapes, without making changes that directly influences climate.

Ultimately, we need to return all the materials and equipment borrowed from our collaboration partners Nordværk. This means we must go back and recycle the materials at the recycle station we borrowed them from. This also made us think further about how the art and sustainability debate can be brought to other aspects of society. We would like to talk to the employees from the recycling stations what they think about loaning artists recyclable materials for their practices. This could even take the project to an entrepreneurial direction. As the recycling stations could either re-sell or loan their materials out to artists or other people who could benefit from this solution. While we were at the recycling station gathering material, we overheard one of the employees say that it was nice how the products could be used, which could indicate that the recycling stations perhaps would find such an entrepreneurial direction positive. We even talked to one Art and Technology student who pointed out that it would make sense to have a collaboration with the recycling station as they could have materials for art students to borrow for their artworks while being students to pursue a greener agenda. Furthermore, it could also be interesting to see what other companies or businesses who could benefit from this solution as well, what purpose could the materials serve.

9. Conclusion

In the report we examined the potential of engaging an audience in discussions about environmental sustainability through dynamic sculptures. Our primary goal was to explore how the design and sensory elements of *Beneath The Weight*, can raise awareness and create intellectual conversations on sustainable production and climate change.

The project shows that dynamic sculptures such as *Beneath The Weight* can effectively unite aesthetic appeals with meaningful messages on sustainability. The research showcased the critical role of material and design choices, in expressing the messages. The theoretical frameworks of Dewy and Jantzen provided insight into the interaction between art, experience and public engagement. These findings suggest that art can be an important medium of promoting awareness of sustainability, serving as a bridge between scientific data and societal viewpoints. Providing an intellectual engagement, the audience is encouraged of deeper reflection and discussion on environmental sustainability, adding discussion to the discourse on sustainability.

Analysing conversations with participants enabled a systematic evaluation of the artwork's impact on the audience. This created a thorough examination of the design, material sustainability and participants' responses gained from the artwork. Through working with

material sustainability, we face challenges of ensuring the movement of the body, being smooth and coherent with the aesthetic appearance. Another challenge of the project included the aesthetic appearance including colour and textures, due to material choices.

Future directions of the artwork includes expanding the project's scope to work within varied cultural and societal contexts, providing a broader understanding of the role of art in relation to sustainability. *Beneath The Weight* works as an example of how art can inspire critical thoughts and discussion of art's possible role in sustainability. Although we still face environmental issues, art can provide a meaningful and powerful tool to encourage and enhance conversations on an environmental and sustainable future.

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Appendix

Appendix 1: Commented_Processing – attached in Digital Exam

Appendix 2: Commented_Arduino_code – attached in Digital Exam

Appendix 3: Walk along interview – attached in Digital Exam